

Table S3. Comparison of linear lactate–pyruvate parameters between "Always LPR > 25" and "Never LPR > 25" cohorts

	Always LPR > 25	Never LPR > 25	Δ^a	P-value^b
Pre-filter % discarded ($m \leq 0$) ^c	0.0% (0.0%, 4.2%)	0.0% (0.0%, 0.0%)	0.0%	—
LPG (m) ^d	27.39 (23.82, 33.78)	14.59 (12.44, 18.20)	12.80	< 0.001
Linear intercept (b , mM) ^d	1.62 (0.58, 2.76)	0.19 (-0.15, 0.37)	1.43	< 0.001
% Type P _b	100.0% (81.3%, 100.0%)	66.3% (51.6%, 85.3%)	33.7%	< 0.001

Values are cohort median (IQR) of patient-level values. Cohort sizes: Always = 39 patients, 82 retained segments; Never = 38 patients, 92 retained segments.

^a Δ = Always – Never.

^bP-values from one-sided Wilcoxon tests (H1: Always > Never) with Benjamini-Hochberg correction.

^cPre-filter only: per-patient percent of datapoints in segments with $m \leq 0$ before filtering. All other rows use retained data ($m > 0$).

^dPatient-level values are segment-weighted means (weights = segment size).

b, linear intercept; **IQR**, interquartile range; **LPG (m)**, lactate–pyruvate gradient; **LPR**, lactate/pyruvate ratio; **P_b**, positive intercept ($b > 0$).